ROBOTiX CLUB

National InstituteofTechnology, Raipur
G. E. Road, Raipur – 492010 (C. G)

ROBOFEST'26

ROBOTHON

About the event:

Prepare for an intense 24-hour sprint of innovation with Robothon, a premier simulation-based hackathon hosted by the ROBOTiX Club. This year, the challenge goes virtual, focusing on five high-impact, ongoing projects: GPS-Denied Drones, Swarm Drones, Underwater Robotics, Nanobots, and Six-Legged Rough Terrain Rovers. Teams of 2–4 members must select their Problem Statement (PS) at the time of registration and begin developing their simulation models immediately. A few days before the main event, progress will be evaluated to shortlist the top 10– 12 teams who will advance to the final stage. During the 24-hour live event, finalists will be tasked with modifying and optimizing their simulations in real-time to meet new technical hurdles. It's a race against the clock where creativity and technical skill meet high-stakes competition, culminating in a showcase of the best-performing simulations and cash prizes for the winners.

Testing & Evaluation Procedure (Applicable for all the problem statement given below):

To ensure fairness, robustness, and real-world adaptability of the developed simulations, the evaluation process will consist of two structured phases:

1. Pre-Event Testing Phase (Shortlisting Round)

- Each team must design and prepare its own testing environment based on the selected Problem Statement.
- The testing environment should realistically reflect the constraints, objectives, and operating conditions of the chosen domain (e.g., obstacle density, terrain roughness, communication limits, hydrodynamic effects, signal attenuation, etc.).
- Teams must clearly document:
  - Environment assumptions
  - Simulation parameters
  - Performance metrics
  - Hardware/software constraints considered
- During the evaluation prior to Robothon, teams will:
  - Demonstrate their simulation running inside their self-designed testing environment.
  - Explain how their environment validates navigation, control stability, obstacle avoidance, coordination, or task execution.
  - Present quantitative performance metrics such as:
    - Task completion time
    - Energy efficiency (if modeled)
    - Collision count
    - Path optimality

- Communication efficiency (for swarm systems)
- Stability metrics (tilt angle, oscillations, etc.)

Based on robustness, realism of the testing setup, technical depth, and performance metrics, the top 10–12 teams will be shortlisted for the final 24-hour Robothon.

2. Final 24-Hour Robothon Testing Phase

During the live 24-hour event, teams will be required to validate their simulation under modified or entirely new testing conditions.

The organizing team may:

- Modify the testing environment created by the team
  (e.g., add dynamic obstacles, increase terrain roughness, introduce communication delay, sensor noise,
  current disturbances, or partial system failures)

OR

- Provide a completely new standardized testing environment prepared by the organizers

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In either case, teams must:

- Adapt and optimize their simulation in real time.
- Tune control parameters, algorithms, or system architecture if necessary.
- Ensure stability and task completion under the new constraints.
- Demonstrate robustness against environmental uncertainty.

Evaluation Criteria During Final Testing

Simulations will be judged based on:

- Adaptability to new or modified environments
- Algorithm robustness and fault tolerance
- Computational efficiency
- Scalability (especially for swarm systems)
- Quality of system architecture
- Clarity of technical explanation
- Reproducibility of results

The final ranking will heavily emphasize how well the system performs outside its originally designed testing conditions, reflecting real-world engineering challenges.

Deliverables (Applicable for all the problem statement given below):

• ROS2/ WeBOT Package
• Upload a complete package to a public GitHub repository.
• Include a detailed README with installation, setup, and execution instructions.
• Demonstration Video
• Technical Presentation (PowerPoint)

Must cover:

• Overall software architecture
• AI/ML algorithms (if any) used for navigation and obstacle avoidance
• Key implementation details and design choices

Outcomes:

Robothon is designed to push the boundaries of virtual prototyping and technical endurance. Participating in this event significantly enhances problem-solving abilities, as teams must navigate complex scenarios within specialized fields like swarm intelligence and underwater navigation. The 24-hour format serves as a crucible for teamwork, requiring seamless collaboration and role delegation to meet tight deadlines. Furthermore, the requirement to modify simulations on the fly during the final round provides invaluable experience in real-time engineering, teaching participants how to iterate and troubleshoot under intense pressure. Ultimately, students walk away with a deep understanding of industry-standard simulation tools and the mental agility required to solve the engineering challenges of tomorrow.

The problem statements will be based on the five ongoing projects currently being undertaken by the ROBOTiX Club. These projects include:

1) GPS-Denied Navigation System
2) Swarm Drones
3) Underwater Robot
4) Nanobot
5) Six-Legged Rough Terrain Rover.

1) GPS-Denied Navigation System:

The project proposes a low-cost, autonomous drone integrating RGB-D vision, 2D LiDAR, and onboard computation (NVIDIA Jetson) to perform real-time SLAM, obstacle avoidance, and path planning without GPS. It bridges the gap between laboratory-level SLAM research and real-world deployment by offering an open, reproducible platform validated through simulation and field testing. Deliverables include a working prototype, datasets, and open-source ROS2 packages, strengthening India's capability in a/utonomous navigation for critical environments.

The problem statements related to this project are:

1. Simulate autonomous navigation of a drone from a start location to a goal location in a 2D environment using the A* algorithm while avoiding obstacles.

2. Simulate a drone using PID controller such that it maintains a certain vertical height and avoids obstacles.

3. Develop an autonomous navigation system for underground or tunnel environments where GPS signals cannot reach, using alternative sensing methods. (Either bot or drone can be used for simulation)

2) Swarm Drones:

The proposed project aims to design and implement a cost-effective swarm of drones capable of cooperative surveillance, mapping, and real-time SLAM in both GPS-enabled and GPS-denied environments. Each drone will possess distinct sensing and computational capabilities, allowing multi-purpose functionality such as garbage monitoring, disaster mapping, and defense reconnaissance. The initiative focuses on building decentralized algorithms for coordination and communication, enabling resilience, adaptability, and efficient data fusion. The expected outcome is a scalable, autonomous swarm UAV platform that can assist in civil, environmental, and defense applications.

The problem statements related to this project are:

1. Lightweight Swarm Navigation in Dynamic Environment (Webots/ROS Simulation)
Desc: The drones fly in coordinated formations like triangle or square to efficiently cover and scan the entire area. This formation-based navigation ensures maximum coverage, better stability, and collision avoidance while moving through the environment.

2. Energy-Efficient Lightweight Swarm Task Allocation (Simulation)
Desc: Multiple drones operate simultaneously and share real-time location data of detected tasks (like garbage spots). Once garbage is detected, the drone marks the exact coordinates and integrates it into a digital map. All drones update this map collaboratively, ensuring faster response and accurate monitoring.

3. Compact Hybrid Lightweight Drone Design (Design Challenge)

3) Underwater Robot:

This project, developed by the ROBOTiX Club at NIT Raipur, focuses on the design and development of a bio-inspired, autonomous underwater robot capable of reliable navigation and communication in subsea environments. Conventional underwater vehicles face issues such as GPS unavailability, sensor drift, signal attenuation, and inefficient energy usage. This project aims to overcome these limitations using an integrated navigation, communication, and safety framework.

The problem statements related to this project are:

1. AUV Sensor Placement Simulation with Demonstration of Blind Spots
Desc: Design and simulate sensor layouts on an underwater robot and visually show regions the sensors cannot see.

2. Simulation-Based Hydrodynamic Force Modeling for an AUV
Desc: Use simulation data to build a simple model showing gravity, drag, thrust, buoyancy forces, etc. acting on an AUV in water.

3. Collision Avoidance for Multiple AUV Simulation
Desc: Coordinate several simulated underwater robots so they move toward their goals without colliding with each other or obstacles.

4) Nanobot:

This project offers a breakthrough in spinal cancer treatment by using magnetically controlled nanoparticles to eliminate residual cancer cells non-invasively. It aims to restore mobility, reduce surgical risks, and improve outcomes for all patients, including the elderly. With biocompatible materials and scalable methods, it supports long-term sustainability and advances public health through safer, more effective cancer care.

The problem statements related to this project are:

1. The Blood-Brain Barrier (BBB) Breach
Task: The robot is placed in a capillary(fluid environment) with varying pore sizes on its wall which are closed. The robot has to navigate toward the correct pore without hitting the wall and has to emit a specific biochemical key (a signal or sequence) to open the correct pore and has to enter in it.

2. Chemotactic Plume Tracking
Task: To mimic a bacteria or nanorobot to find target by following a chemical trail. In a 3D space, there is a source which is releasing a biomarker that follows a Gaussian distribution (it gets weaker as you moves away). The nanorobot doesn't have the map; it only knows the concentration of chemical at its current position. It must find the source in shortest time possible using some gradient-climbing algorithm.

3. Swarm Coordination for Structural Repair
Task: Using a swarm of smaller units to fix a larger structure. A virtual pipe (blood vessel) has multiple cracks in it. A swarm of 5-10 virtual nanorobots must navigate to the target and coordinate their movements to cluster around the crack simultaneously the bridge the gap. Bots can only talk to each other if they are within a certain distance, requiring them to stay in a chain or mesh to pass information.

5) Six-Legged Rough Terrain Rover:

This project proposes an iterative, tightly integrated hardware–software development cycle for building a six-legged, rugged terrain robot with advanced navigation and modular sensing capabilities. The overarching goal is to create a versatile platform that blends affordability with advanced autonomy, ensuring practical deployment in real-world conditions where cost, reliability, and adaptability are equally critical. To achieve this, the project is divided into staged phases—starting from simulation and mechanical design to hardware prototyping, perception integration, and field validation. This staged approach minimizes risks, accelerates iteration cycles, and ensures that all major subsystems are validated both independently and collectively before full-scale deployment.

The problem statements related to this project are:

1. Design and simulate a six-legged rover or Quadruped robot in ROS2 that autonomously navigates through a rough terrain environment (rocks, slopes, uneven ground) while maintaining stability. The robot must monitor its body tilt and adjust gait to prevent tipping. (Simulation)

2. Develop a ROS2-based simulation of a quadruped or six-legged robot that autonomously explores an indoor room environment, generates a map using onboard sensors, detects objects present in the room, and counts the total number of objects. The robot must localize detected objects on the map and output a final report containing the object count and their approximate positions. (Simulation)

3. Develop a ROS2-based simulation of a quadruped or six-legged robot that can be manually controlled using keyboard teleoperation while simultaneously building a map of the environment and detecting objects using onboard sensors. The robot must visualize the generated map, identify objects during operation, and log their locations for later analysis. (Simulation)

4. Optimized Leg Architecture Design (Simulation-Based Mechanical Design)